

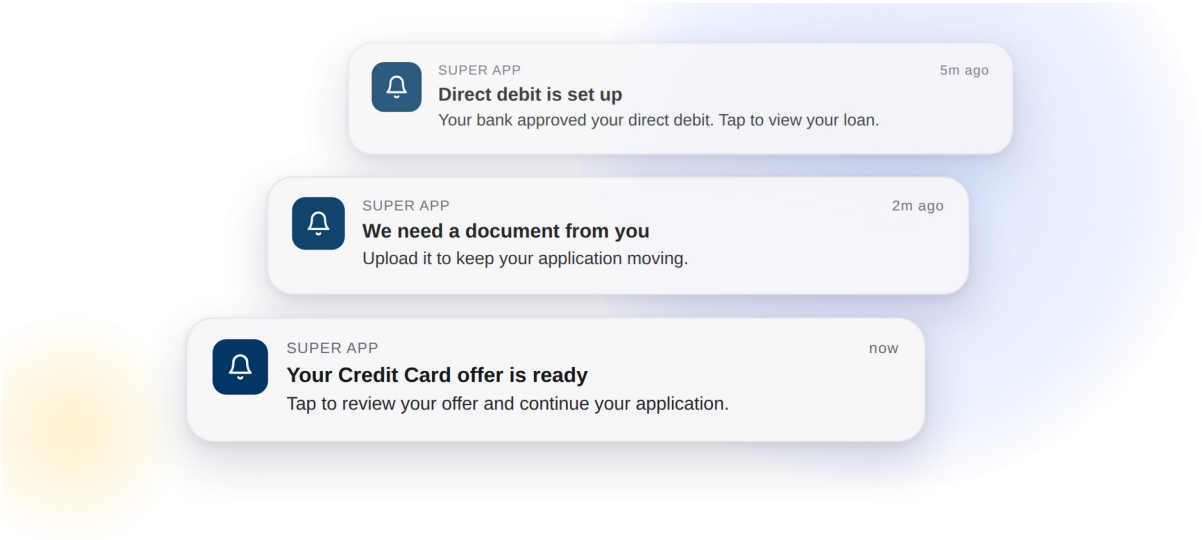


BUSINESS REQUIREMENTS DOCUMENT

Push Notification

Customer journey alerts in the Super App — Credit Card, Personal Loan and CASA

V1.4 / 08 October 2026



VERSION HISTORY

Date	Version	Author	Change Description
28-09-2026	1.0	Hailey (Appro)	Created BRD
28-09-2026	1.1	Hailey (Appro)	Trigger events and push content in one table; push results tracked in the backend only, not on the audit trail; sample request and response exactly as the middleware specification
06-10-2026	1.2	Hailey (Appro)	Business review of 29/09 incorporated: updated wording for P01, P04, P05 and P09; device and session scenarios; reminder stop rules and sending time; middleware specification v0.2 values and error codes (05/10), retry values to be confirmed by Business; duplicate-prevention rule BR5; future enhancements (post-MVP, dedicated development team)
07-10-2026	1.3	Hailey (Appro)	Business-approved event content of 07/10: Arabic title and body per notification (new section 3.2); P09, P10 and P11 wording per the approval; resuming screen per notification and scope updates; automatic retry capped at a maximum of 10 attempts (interval remains a backend parameter); integration items tracked with Reem Bank IT and Avanza outside the BRD
08-10-2026	1.4	Hailey (Appro)	Baseline version for MVP 1.1 development — Business approval to proceed received (Head of Retail, 08/10). No content change from V1.3.

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1. FEATURE OVERVIEW AND KEY COVERED AREAS

This document describes the requirements for **Push Notification** in the Reem Bank customer journey for **Credit Card, Personal Loan and CASA**, on iOS and Android.

Today the customer learns about decisions and required actions on an application — a credit decision, a Direct Debit Authority response, the end of the cooling-off period — by **email and in-app notification only**. When the customer is outside the Super App, nothing reaches the device, and the application waits until the customer returns.

Push Notification adds a device-level alert at each agreed trigger event. The Apro platform sends the push request to the Reem Bank middleware Push API, following the same integration pattern as the existing SMS and email integration. When the customer taps the notification, he will be navigated to Super App Login screen and the Apro journey opens at the screen where the application continues (Resume screen) or Super App depending on the events. (Refer to the below **Section 3.1. Trigger events and push content** for further details)

This BRD covers the following areas:

- **Trigger events and push content** — nine notifications, each with its title and body in English and Arabic, and the screen it opens
- **Language** — English or Arabic, following the customer's current language
- **Middleware integration** — the Push API request, headers and response
- **Response handling and automatic retry**
- **Business rules** — personal data and excluded templates

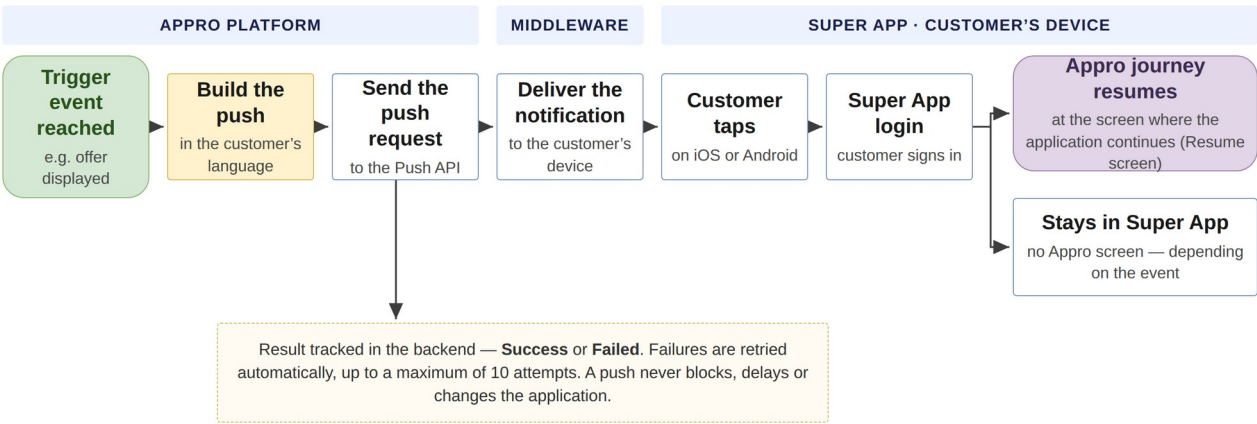


Figure 1 — How a push reaches the customer (green = start, amber = system action, blue = step, purple = end)

Key steps

1. A trigger event is reached on the application, for example an offer is displayed or a document is requested.
2. The system builds one push for the event: the Push ID, the title and the body, in the customer's current language.
3. The system sends the push request to the Reem Bank middleware Push API.
4. The middleware delivers the notification to the customer's device through the Super App.
5. The customer taps the notification, and the Apro journey opens at the screen for that event.
6. The result is tracked in the backend. A failed push is retried automatically and never blocks the application.

2. SCOPE

2.1 Product scope

Credit Card, Personal Loan and CASA (now) and **Auto Loan/Mortgage Loan (Future)** on **iOS and Android**. Each notification applies only to the products listed against it in section 3.

2.2 In-scope notifications

- Refer to the below Section 3.1

2.3 Not in scope

- OTP messages — never sent as push (BR3).
- Bank and internal-user notifications — never sent as push (BR4).
- Configuring push content in Super Portal — content is fixed per notification in this release.
- Tracking customer actions on a notification — open, dismiss or ignore.
- Changes to the existing email and in-app notifications — To be enhanced in later stage

3. TRIGGER POINTS AND PUSH CONTENT

3.1 Trigger events and push content

A push is fired at the same trigger point as the existing email and in-app notification for the same event. Existing email and in-app notifications remain unchanged.

Push ID	Resuming screen	Applies to	Trigger event	Title (EN, max 40 characters)	Body (EN, max 100 characters)
P01	AIP screen (Appro SDK)	CC · PL	AIP given (at least one offer displayed), or Credit Queue approves / overrides the application	Your offer is ready	Review your offer and continue your application.
P01 (CASA)	AIP screen (Appro SDK)	CASA	Account application approved in principle	Your account is approved	Tap to continue and complete your account opening.
P02	AIP screen (Appro SDK)	CC · PL · CASA	Scheduled reminder once per day for 15 days, starting after the first offer is displayed	Your offer expires in %%COUNTING_DOWN%% days	Complete your application before your approval expires.
P06	KFS screen (Appro SDK)	CC · PL	Cooling-off end date reached (start + 5 days, configurable) with no customer action; application returns to Approved In Principle	Your application is active again	Your cooling-off period has ended. Tap to continue or cancel.
P09	Super App	PL	Customer's bank rejects the Direct Debit Authority	Your direct debit needs attention	Your bank could not approve the request. Contact Us.
P04	Application Unsuccessful screen (Appro SDK)	CC · PL · CASA	Queue user rejects (Credit, Sale, Risk or Compliance), Rule Engine filtration fails, or minimum income not met	There's an update on your application	Tap to view the latest status of your application.
P10	PL Congratulation screen (Appro SDK)	PL	Disbursement Checker approves and releases the disbursement	Your loan is on its way	Funds will reach your account shortly.
P11	AIP screen (Appro SDK)	CC · PL · CASA	Card issued (CC, CASA) or loan approved (PL)	Your %%PRODUCT_TYPE%% is ready	Congratulations Tap to see what happens next.
P03	Super App	CC · PL · CASA	Offer validity (30 days) elapses with no acceptance	Your offer has expired	You can reapply whenever you're ready.

3.2 Arabic content — approved by RB Business (07/10)

The Arabic title and body below were approved by the Business together with the English content. The push follows the customer's current App language (section 3.4).

Push ID	Title (AR, max 40 characters)	Body (AR, max 100 characters)
P01	تمت الموافقة المبدئية على طلبكم	يُرجى الاطلاع على التفاصيل لمتابعة استكمال طلبكم.
P01 (CASA)	تمت الموافقة على طلب فتح حسابكم	يُرجى النقر لاستكمال الإجراءات المتبقية لفتح الحساب.
P02	تنتهي صلاحية التمويل المقترح خلال %%COUNTING_DOWN%% أيام	يُرجى استكمال طلبكم قبل انتهاء صلاحية الموافقة الممنوحة لكم.
P06	تمت إعادة تفعيل طلبكم	انتهت فترة التراجع. يُرجى النقر للمتابعة أو إلغاء الطلب.
P09	يتطلب الخصم المباشر إجراءً منكم	تعدّ اعتماد طلب الخصم المباشر من قبل بنككم. يُرجى مراجعته البنك.
P04	يوجد تحديث على طلبكم	يُرجى النقر للاطلاع على آخر مستجدات طلبكم.
P10	جارٍ تحويل مبلغ القرض	سيتم إيداع المبلغ في حسابكم قريباً.
P11	%%PRODUCT_TYPE%% جاهز الآن	تهانينا، يُرجى النقر للاطلاع على الخطوات التالية.
P03	انتهت مدة صلاحية الموافقة	يسعدنا استقبال طلبكم مجدداً في الوقت الذي يناسبكم.

3.3 Content rules

- [%%PRODUCT_TYPE%%] and [%%COUNTING_DOWN%%] must be replaced with actual values before sending. No placeholder is ever shown to the customer.
- The title must not exceed **40 characters** and the body **100 characters**, after replacement.
- Content is fixed per Push ID and is not configurable in Super Portal in this release.

3.4 Language

- The push is sent in **English or Arabic**, following the language the customer is currently using.
- Arabic title and body for every notification — approved by RB Business on 07/10 (section 3.2).

3.5 Device and session scenarios

Each scenario has one owner. App rows are covered by this BRD; the Super App and middleware rows are owned by Avanza.

Scenario	Owner	Behaviour
Customer logged out or session expired	Super App	The Super App completes the login first, then opens the Appro journey at the screen for the event.
Customer changed device, or multiple registered devices	Super App	The push follows the device-token registration held by the Super App.
Push permission disabled on the device	Super App	No device alert. The application and the existing email and in-app notifications continue unchanged.
Device token invalid or expired	Middleware	The middleware returns 404 UNREGISTERED — tracked in the backend as Failed, no retry (section 4.5).
Application completed, cancelled, withdrawn, rejected or expired	Appro	The outcome notification for the event itself is sent once (e.g. P03, P04, P11). After the terminal status, no further push is sent and the P02 reminders stop immediately (section 3.6). On a late tap, the customer is navigated to the Super App screen.

Scenario	Owner	Behaviour
Customer no longer eligible	Appro	A push is built only at its trigger event on a live application. On a tap, the customer is navigated to the Super App screen.

3.6 Reminder and stop rules (P02)

- P02 is sent once per day for 15 days, starting after the first offer is displayed. The interval and the maximum count are backend configuration parameters (default: daily, 15).
- P02 stops immediately when the customer selects or rejects the offer, or when the application is completed, cancelled or withdrawn, rejected or expired, or the product is booked or issued.
- One push per Push ID per trigger occurrence per application (BR5).

3.7 Sending time

- Event-driven pushes are sent immediately at the trigger event.
- Scheduled pushes — the P02 reminder and P03 (offer expired) — are released at 10:00 UAE time (backend configuration parameter).

4. PUSH NOTIFICATION API INTEGRATION

4.1 Request details

Item	Value
Method	POST
Endpoint	http://<MW URL>/api/notifications/push
Authentication	Bearer token

4.2 Header parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
stan	String	Y	Unique system audit trace number in each request	Generated by Appro, unique per request. 6–12 characters (specification v0.2).
Authorization	String	Y	Authorization header	Bearer <token>
Content-Type	String	Y	Request format	application/json
channel_id	String	Y	Channel id of the calling system	value assigned to Appro

4.3 Request parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
requestId	String	Y	Unique message id	Generated by Appro, unique per push. Every retry reuses the same requestId.
timestamp	String	Y	Request date-time	ISO 8601 UTC with milliseconds, e.g. 2026-09-28T06:49:02.366Z
type	String	Y	Notification type	Fixed: APPRO_NOTIFICATION

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
action.id	String	Y	Push identifier	Push ID from section 3, e.g. P01. P01 (CASA) is sent as P01 with action.value = CASA
action.type	String	Y	NONE / POPUP / APPRO_JOURNEY	APPRO_JOURNEY: a tap opens the Appro journey at the screen for the event. NONE: no forced action — the tap opens the Super App; used for P03 and P09 (per the alignment call of 05/10). POPUP: not used by Appro.
action.value	String	Y	Product	CREDIT_CARD / PERSONAL_LOAN / CASA, from the application.
recipient.mobilePhone	String	Y	Customer mobile number	From the application. Format 9715XXXXXXX — country code 971, no plus sign (specification v0.2).
recipient.customerId	String	Conditional	Customer CIF — required if ETB	ETB: customer CIF. NTB: not sent.
recipient.email	String	N	Customer email	No mapping – Optional field
content.title	String	Y	Push title	Per section 3.1, in the customer's current language
content.body	String	Y	Push body	Per section 3.1, in the customer's current language
content.message	String	N	Notification message	Sent empty (""). Pop-up content, used only with action type POPUP (specification v0.2).
content.language	String	Y	Language — "EN" / "AR"	Customer's current language — "EN" or "AR" (specification v0.2).

4.4 Response parameters

Parameter	Type	Description	Values / Handling
messageId	String	Message id returned by the middleware	Tracked in the backend
error.code	String	Result code	"000" = Processed Ok. Full error-code list in section 4.5.
error.message	String	Result description	Tracked in the backend with error.code

4.5 Error codes (middleware specification v0.2)

Code	Description	Appro handling
000	PROCESSED OK	Push sent — tracked in the backend as Success.
403	SENDER_ID_MISMATCH	Final, no retry — configuration issue, raised with Avanza. Tracked as Failed.
404	UNREGISTERED	Final, no retry — the device token is not registered. Tracked as Failed.

Code	Description	Appro handling
429	QUOTA_EXCEEDED	Retried per section 5.
500	INTERNAL SERVER ERROR	Retried per section 5.

5. RESPONSE HANDLING AND RETRY

- If the HTTP code is **20x** and [error.code] = **"000"** (Processed Ok) → the push is sent → tracked in the backend as **Success** → the journey proceeds.
- Else → the system retries automatically after **X minutes**, up to a maximum of **10 attempts**. The retry interval is a backend configuration parameter.
 - "Else" covers a timeout, any non-20x HTTP code and any [error.code] other than "000".
 - Error codes 429 and 500, and a timeout, are retried. 403 and 404 are final and are not retried (section 4.5).
 - Every retry reuses the same requestId, so a retry never creates a duplicate notification (BR5).
- If the push is still not successful after the last attempt → tracked in the backend as **Failed** → the journey proceeds. A push failure must not block, delay or change the application status.
- The system tracks the API delivery result only, in the backend; it is not shown in the Super Portal audit trail. Customer actions on the notification (open, dismiss, ignore) are not tracked. The backend record per push: Push ID, date and time, number of attempts, final status (Success / Failed) and the failure reason. Until a Super Portal view is available (section 8), Appro support provides the status of a push for a given application on request from the Contact Centre.

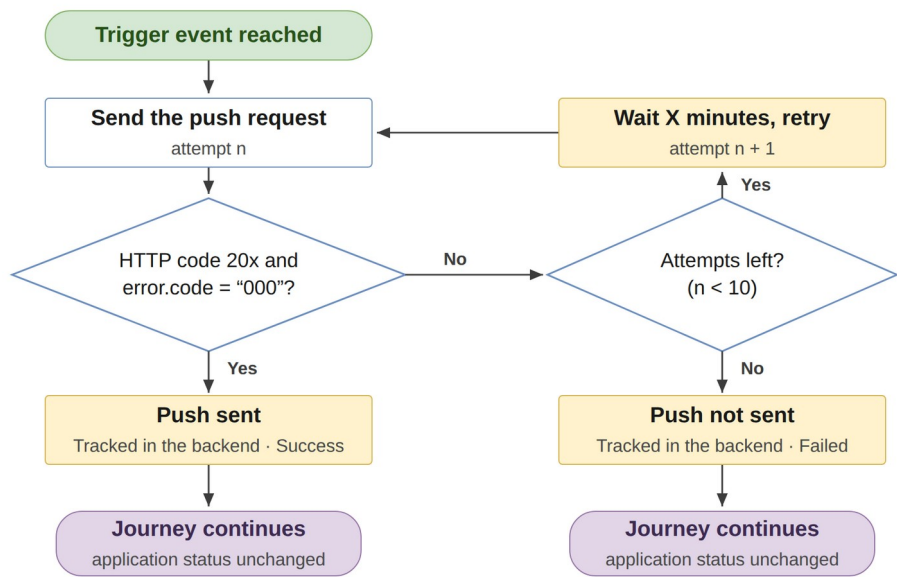
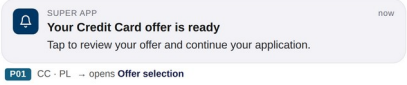
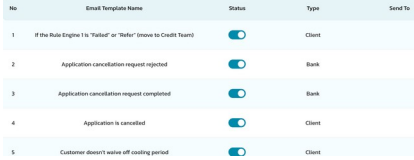
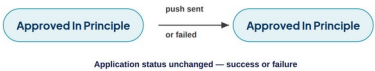


Figure 2 — Response handling and automatic retry (automatic retry — maximum 10 attempts; interval configurable)

6. BUSINESS RULES

No.	Rule
BR1	Push is additional; existing email and in-app notifications are unchanged.
BR2	No personal data in the title or body: no name, Application ID, IBAN or amounts.
BR3	OTP content (ET17, ET18, ET35, NT04, NT05) is never sent as push.
BR4	Bank and internal-user templates are never sent as push.
BR5	One notification per event: a push is sent once per Push ID per trigger occurrence per application, and every retry reuses the same requestId — a system retry, timeout or reprocessing never creates a duplicate.

7. IMPACT ANALYSIS

Area	Impact	Screen
Customer journey (Super App)	A device notification at nine trigger events. A tap opens the Appro journey at the screen for that event; existing journey screens are reused as the destinations.	
Communication Setup	No change. Existing email and in-app templates keep working as today and remain the trigger reference for each push. Push content is fixed per notification in this release and is not added to Communication Setup.	
Audit trail	No change in this release. Push results are tracked in the backend only; Appro support provides the status of a push on request (section 5). A Super Portal view is a post-MVP enhancement (section 8).	No screen change
Application status	No change. A push — sent or not — never changes, blocks or delays the application.	
Reporting	The API delivery result is tracked. Customer actions on a notification (open, dismiss, ignore) are not tracked in this implementation scope	No screen change

8. FUTURE ENHANCEMENTS (POST-MVP)

The items below are agreed as enhancements after the MVP 1.1 sign-off, delivered by the dedicated development team based on launch feedback — not as change requests. The MVP design is scalable, so they are added on the same framework, without redesign.

- Business-managed configuration in Super Portal, wherever technically feasible without source-code modification: enable / disable per notification, English and Arabic content, reminder frequency and maximum count, validity period, priority, trigger and product mapping, and new Push IDs.
- Sending notifications from Super Portal.
- Push visibility in Super Portal: an audit-trail entry or a notification report with the sent / failed status per application.
- Configurable sending hours and customer local time.
- Additional products, journeys and notification types on the same framework.

APPENDIX 1: SAMPLE REQUEST/RESPONSE

Sample request

```
{
  "requestId": "1234567891011121",
  "timestamp": "2020-07-20T06:49:02.366Z",
  "type": "APPRO_NOTIFICATION",
  "action": {
    "id": "P01",
    "type": "APPRO_JOURNEY",
    "value": "CASA"
  },
  "recipient": {
    "mobilePhone": "971528045456 ",
    "customerId": "104278",
    "email": "abc@test.com"
  },
  "content": {
    "title": "CASA Journey",
    "body": "Please resume your CASA journey",
    "message": "",
    "language": "EN"
  }
}
```

Sample response

```
{
  "messageId": "0:1790341509514173%d5881a7ad5881a7a",
  "error": {
    "code": "000",
    "message": "Processed 0k"
  }
}
```

Product mapping

Product on the application	action.value
Credit Card	CREDIT_CARD
Personal Loan	PERSONAL_LOAN
CASA	CASA